LSTV Reviewer

3D Slicer module for streamlined LSTV landmark annotation

> **Delivery objective**
>
> The CTSpinoPelvic1K dataset and the segmentation model trained from it are intended for direct deployment as **3D Slicer's default spinal-CT segmentation backend**, replacing TotalSegmentator for spinopelvic regions. Your annotations train both the LSTV-aware segmentation model and an automated spinopelvic-parameter (PI / PT / SS / LL) measurement module that ships in the same Slicer extension.

What this module does

Reduces per-case annotation to three button clicks. The module loads the CT, segmentation masks, and landmark template automatically, saves the result with the correct filename in the correct directory, and tracks completion progress across all 33 cases.

Installation

About 2 minutes; no terminal, no programming. You will need **3D Slicer 5.x or newer** (free at slicer.org) and the **LSTVReviewer.py** file provided with the dataset.

1. Save **LSTVReviewer.py** into a new folder anywhere on your computer. The folder name does not matter, but pick a location you will not move or delete. Examples:
   ```
   macOS / Linux:  ~/lstv-reviewer/LSTVReviewer.py
   Windows:          C:\Users\you\lstv-reviewer\LSTVReviewer.py
   ```
2. The folder must contain **only** this file. The filename must be exactly **LSTVReviewer.py** (case-sensitive).
3. Open 3D Slicer.
4. From the menu bar, choose `Edit` → `Application Settings`.
5. In the left panel of the Settings window, click **Modules**.
6. Scroll to **Additional module paths**.
7. Click the green **Add** button (the '+' icon).
8. In the file browser that appears, navigate to and select the **folder containing LSTVReviewer.py** (the folder, not the .py file). Click **Open** or **Select Folder**.
9. Click **OK** at the bottom of the Application Settings window.
10. When prompted, click **Yes** to restart Slicer. If no prompt appears, quit Slicer and reopen manually.
11. After Slicer restarts, find the **Modules** dropdown in the top-left toolbar. Under the **Annotation** category, click **LSTV Reviewer**. The module panel appears.

Daily usage

1. Click **Browse...** and select the `lstv_review/` folder you received. The cases dropdown populates.
2. Pick a case from the dropdown (■ = pending, ✓ = saved).
3. Click **Load Selected Case**. CT, masks, and landmark template all load. The view auto-switches to the Markups module.
4. Place each fiducial: click a label in the Control Points list, press `Ctrl+Shift+A`, then click on the anatomical location in any slice viewer. Slicer auto-advances to the next fiducial.
5. For absent landmarks (e.g., S5 foramen in a 4-segment sacrum, or fusion bridge in a non-fused case): right-click the fiducial in the list → *Unset position*. It remains in 'preview' status and is treated as absent.
6. Switch back to `Modules → Annotation → LSTV Reviewer`. Click **Save Landmarks**. The file is written as `landmarks.mrk.json` in the case directory automatically.
7. Click **Open review.txt**. Fill in the brief categorical form in your default text editor, save, close.
8. Click **Next incomplete**. Repeat.

What is generated automatically

You never type JSON by hand. Three things are written for you:

Output	Generated by
landmarks.mrk.json	3D Slicer's Markups module on Save
Castellvi grade (Ia / Ib / IIa / ...)	Server-side post-processing from your placements
Mahato spectrum grade (B / C / D / E)	Server-side post-processing from your placements
All linear / angular morphometrics	Server-side post-processing from coordinates

*The **only** file you edit by hand is `review.txt`, which is plain text (a few Field: value lines).*

Troubleshooting

Symptom	Fix
Module does not appear after restart	Check filename is exactly LSTVReviewer.py (case-sensitive). Confirm in Edit → Application Settings → Modules that the path appears in 'Additional module paths'. The path must be the folder, not the .py file.
'Folder does not contain expected LSTV subtypes'	You selected the wrong folder. Browse to the lstv_review/ folder containing subtype subdirectories (sacralization/, lumb/, etc.).
Save Landmarks does nothing	A case must be loaded first. Status panel should read 'Loaded: ...' before saving works.
Open review.txt does not open editor	Some Linux desktops do not associate .txt with an editor. Open the file manually from your file manager.
I want to change a placed landmark	In the Markups Control Points list, right-click the fiducial → 'Unset position'. Then re-place it by clicking again in the slice viewer.

Authorship. Co-authorship is offered to reviewers who complete subtype confirmation and place the 26 required landmarks for each LSTV case. The 5 optional spinopelvic landmarks (femoral heads, L1 endplate, S1 endplate posterior) contribute to a follow-up paper on automated spinopelvic measurement, with continuing co-authorship. Questions: Gregory Schwing, Wayne State University, Department of Neurosurgery.